Geodesic: a parallel test-particle geodesic tracker for the Einstein Toolkit

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Abstract

We present `Geodesic`, a new thorn for the Einstein Toolkit that integrates test particles on geodesics of a spacetime in Kerr–Schild Cartesian coordinates. The thorn provides two modes of metric evaluation: an exact analytic Kerr–Schild metric, independent of the computational grid and verified symbolically to be a vacuum solution, and a grid mode that reads the metric from the simulation through one of four local samplers selected by a single parameter (`l27`, `local27`, `aei`, `carpet`). Each particle is an independent unit of work, so the particle loop parallelizes trivially and scales to 10^4 particles (a measured 9.2× speedup on the 88 threads of a 44-core node); the same independence underlies an owner-computes MPI mode with the replicated state bit-identical on every rank, and the samplers run unchanged on Carpet AMR meshes. The geodesic equation is integrated with an adaptive Runge–Kutta–Fehlberg (RKF45) scheme and a per-step renormalization of the four-velocity. We verify the thorn by (i) conservation of the energy E, the axial angular momentum L_z and the norm to $\sim 10^{-9}$ over $T = 1000\,M$ on three bound orbits; (ii) grid convergence of the default sampler with a measured local order of 5.6; (iii) a comparison of the grid samplers on a precessing orbit of 100 particles; (iv) stable orbits on a 3-level AMR mesh; and (v) robustness on the two-puncture `GW150914` background: geodesics integrate stably on the `ML_BSSN`-evolved (time-dependent) merger spacetime, and on the frozen initial data the strong-field drops of a near ring are deterministic and essentially back-end-independent. Embedded in a full `GRHydro` evolution, the integration of 10^4 particles adds only a few percent to the total wall time. The thorn is released under the GPLv2+ license together with the configuration files used in this work.

1 Introduction

Test-particle geodesics are a basic diagnostic in numerical relativity and in modelling of accretion flows around black holes [Poisson, 2004]: they provide reference orbits against which self-consistent hydrodynamic and electromagnetic solutions can be compared [Blandford and Znajek, 1977], they define the orbital structure used to interpret line profiles and disk precession [Bardeen, 1973, Gralla and Lupsasca, 2020], and they serve as a validation tool for the spacetime produced by a simulation [Kerr, 1963, Bardeen et al., 1972, Wilkins, 1972]. General-purpose geodesic integrators for 3+1-split spacetimes have been developed for test-particle diagnostics in GRMHD simulations [Bacchini et al., 2018, 2019]; dedicated packages compute light propagation and optical observables on numerical-relativity output [Grasso, 2022] and polarized ray tracing on prescribed metrics [Aimar et al., 2024];

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timelike and plunging Kerr geodesics are available analytically or through public packages [Park and Nasipak, 2024, Dyson and van de Meent, 2023]; and in self-force work the particle worldline is advanced together with the field on the numerical grid [Vretinaris et al., 2026]. None of these, however, is a test-particle geodesic tracker inside the Cactus schedule: the verification studies of Bacchini et al. [2018, 2019] target fixed or given metrics, and none of these codes addresses particle tracking on an evolved spacetime resolved on an AMR mesh distributed across MPI ranks. The standard frameworks for general-relativistic simulations of compact objects, the Cactus code framework [Cactus Code, 2024] and its extension, the Einstein Toolkit [Löffler et al., 2012], do not ship a geodesic tracker for test particles in Kerr–Schild coordinates, which is the coordinate system used by most Kerr initial-data and spacetime-evolution thorns. Users who need geodesics today must either write their own integrator or post-process the evolved metric fields, and in both cases the cost of a global interpolation of the metric at every particle position prevents the straightforward parallel scaling that the problem structure allows.

`Geodesic` is a thorn for the Einstein Toolkit that supplies this functionality. Its design goals are (i) to support both an exact analytic Kerr–Schild metric and a metric interpolated from the simulation grid, so that the same particle set can be tracked on the analytic background or on an evolved spacetime; (ii) to keep the per-particle work completely local, so that the particle loop can be parallelized with OpenMP without any inter-particle communication; and (iii) to be a clean, auditable reference implementation of the geodesic equation in Kerr–Schild Cartesian coordinates, verified against an independent implementation.

The main contributions of this paper are:

1. a thorn that integrates the geodesic equation in Kerr–Schild Cartesian coordinates with an adaptive RKF45 scheme and a per-step four-velocity renormalization;

2. an exact analytic Kerr–Schild metric whose vacuum character is verified symbolically (Sec. 3.1);

3. four local grid-metric samplers selected by one parameter (`l27`, `local27`, `aei`, `carpet`; Sec. 3.2), in which every particle is an independent unit of work, validated by a strong-scaling run with 10^4 particles (Sec. 6);

4. a verification suite: conservation of E, L_z and the norm over $T = 1000\,M$ for three distinct bound orbits, a grid-convergence study showing a measured local convergence order of 5.6 for the default sampler, a four-way comparison of the exact metric against the grid samplers on a precessing orbit of 100 particles, and a robustness run on the two-puncture GW150914 merger background (Sec. 5.6).

The remainder of the paper is organized as follows: Sec. 2 describes the geodesic equation in the 3+1 formulation and the initialization of the particle four-velocity; Sec. 3 the exact mode and the four grid-metric samplers; Sec. 4 the implementation, including the OpenMP and MPI parallelization, the AMR level handling and the particle-lifecycle management; Sec. 5 the verification results; Sec. 6 the strong-scaling results; and Sec. 7 concludes.

2 Method

2.1 Geodesic equation

We work in Kerr–Schild Cartesian coordinates (t, x, y, z) with spin axis along $+z$, in which the Kerr metric of mass M and dimensionless spin a takes the form [Kerr and Schild, 1965]

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2Mr^3}{D}\, l_\mu l_\nu, \qquad l_\mu = \left(1,\, \frac{ay + rx}{Q},\, \frac{ry - ax}{Q},\, \frac{z}{r}\right), \tag{1}$$

with $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$, $Q = r^2 + a^2$, $D = r^4 + a^2 z^2$, and r the Kerr radial coordinate defined by $r^2 = (x^2 + y^2 + z^2 - a^2 + \sqrt{(x^2 + y^2 + z^2 - a^2)^2 + 4a^2 z^2})/2$. The vector l_μ is null, $g_{\mu\nu} l^\mu l^\nu = 0$, and the metric is a vacuum solution, $R_{\mu\nu} = 0$ (Sec. 3.1).

The four-velocity $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$ of a massive test particle obeys the geodesic equation

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = -\Gamma^\mu_{\alpha\beta}\, u^\alpha u^\beta, \qquad g_{\mu\nu} u^\mu u^\nu = -1, \tag{2}$$

where $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols of $g_{\mu\nu}$. Because the metric is stationary and axisymmetric, two quantities are conserved along the geodesic: the specific energy $E = -u_t$ (Killing vector ∂_t) and the axial angular momentum $L_z = -y\, u_x + x\, u_y$ (Killing vector $-y\partial_x + x\partial_y$), with the covariant components $u_\mu = g_{\mu\nu} u^\nu$. Both are used as verification monitors.

2.2 Integration in proper time

The thorn integrates Eq. (2) in proper time τ with an adaptive fifth-order Runge–Kutta–Fehlberg scheme (RKF45, Fehlberg 1969; the `rkf45` driver of the GNU Scientific Library [GNU Scientific Library, 2026]), with an absolute error tolerance of 10^{-8} and zero relative tolerance. Each Cactus time step advances the particle until its coordinate time reaches $t + \Delta t$: the proper-time increment is predicted so that t lands on the target, and any overshoot is removed by a linear back-interpolation, so that all particles share the Cactus time axis. The number of proper-time sub-steps within one Cactus step is bounded by a per-particle cap (`max_iter_cap`, default 2000), so that a particle whose proper-time step has collapsed (as near a horizon) cannot stall the evolution. At the end of every Cactus time step the thorn renormalizes the four-velocity so that $g_{\mu\nu} u^\mu u^\nu = -1$ holds to roundoff, which removes the secular drift of the norm that pure adaptive integrators accumulate. The spatial components u^i produced by the ODE step are kept as they are, and the constraint is solved for u^t as a quadratic,

$$g_{00}\,(u^t)^2 + 2\, g_{0i}\, u^i\, u^t + \left(1 + \gamma_{ij} u^i u^j\right) = 0, \tag{3}$$

selecting the future-directed root. Outside the ergosphere ($g_{00} < 0$) the two roots have opposite signs and the unique positive root is unambiguous; inside the ergosphere the same u^i can in fact correspond to two future-directed solutions (a geometric effect of frame dragging), and the root is chosen by continuity, i.e. the one closest to the u^t produced by the ODE step before renormalization.

2.3 Initialization of the four-velocity

The particle initial conditions are specified in the par file as a position (x, y, z) and the contravariant four-velocity components $u^\mu = (u^t, u^x, u^y, u^z)$ in Kerr–Schild coordinates. At the first time step the thorn evaluates the metric at the initial position and checks the normalization $g_{\mu\nu} u^\mu u^\nu = -1$. If it holds to a tolerance of 10^{-6}, the input is used unchanged and the log says so; otherwise u^t is

re-solved from Eq. (3) with u^i held fixed, and the correction (before/after values) is written to the log. The check is therefore idempotent: feeding the values logged at initialization back into the par file changes nothing. The 10^{-6} tolerance is loose on purpose; the per-step renormalization described above restores the constraint to roundoff on the first step regardless. All orbits used in this paper are generated by first integrating the exact-metric geodesic equation with an independent Python implementation (RK4, $\Delta t = 0.02\,M$) and then writing the resulting exact four-velocity directly into the par file; the initial data are listed in Table 1.

3 Metric evaluation

3.1 Exact Kerr–Schild mode

With `Exact = yes` the thorn evaluates the Kerr–Schild metric and its coordinate derivatives analytically at the particle position (Eq. (1)), so that the result is independent of the computational grid. The analytic Christoffel symbols are obtained from the analytic first derivatives of the metric; no finite differences enter the evolution. As a self-consistency check we verify the vacuum character of the implemented metric symbolically with SymPy [Meurer et al., 2017]: the components of the thorn's metric are checked term-by-term against the Kerr–Schild form Eq. (1) (which also proves l_μ is null), and the Ricci tensor $R_{\mu\nu}$ is assembled from the analytic metric and its analytic inverse $g^{\mu\nu} = \eta^{\mu\nu} - H\,l^\mu l^\nu$ and evaluated at five generic points to 40-digit precision, giving $\max|R_{\mu\nu}| < 10^{-170}$, i.e. the implemented metric is a vacuum solution to beyond machine precision.

3.2 Grid-metric samplers

With `Exact = no` the metric is taken from the simulation grid: the thorn reads the `ADMBase` lapse, shift and spatial metric (declared with schedule `READS` clauses), assembles the covariant metric $g_{\mu\nu}$ at the stencil grid points surrounding the particle, and obtains the spatial metric derivatives there by central finite differences (sixth order in the stencil-based samplers). The Christoffel symbols also require the time derivative $\partial_t g_{\mu\nu}$; how it — and the metric value at the particle's coordinate time — is obtained from the stored `ADMBase` time levels is fixed per sampler below. A stationary metric is detected automatically (the oldest two stored levels agree to relative 10^{-12} on a subsample), in which case $\partial_t g_{\mu\nu} = 0$ exactly and the time differencing is skipped. Within one particle step the connection is frozen at the grid's current time, so for a time-dependent metric the particle step is capped at the grid time step; time-dependent runs therefore need the `ADMBase` fields at three time levels (`Carpet::init_fill_timelevels = yes`). How the metric and the Christoffel symbols are sampled at the (generally off-grid) particle position is selected by the parameter `metric_interp`; the four choices are described below. All four samplers are local: the computation for one particle reads only the grid cells around it and writes no data used by any other particle.

127. The simplest sampler: for a given particle the thorn identifies the 27 grid points of the 3×3×3 cell stencil surrounding it and, for each metric component and each Christoffel symbol, fits the unique polynomial that is at most quadratic in each coordinate and interpolates the 27 stencil values — a 27-term tensor-product Lagrange interpolant [Berrut and Trefethen, 2004]. The fit is a single matrix–vector product with a precomputed 27×27 inverse Vandermonde matrix, and fields that are constant on the stencil collapse to their constant value without the product. The nodal Christoffel symbols are computed from the sixth-order finite-difference metric derivatives, and $\partial_t g_{\mu\nu}$ from a second-order backward difference $(3L_0 - 4L_1 + L_2)/(2\,\Delta t)$ over the three stored time levels

(zero for a stationary metric); the metric values are taken from the current time level only. The fit is cached per particle: it is built once per Cactus time step (and re-evaluated within the step only if the particle crosses into an adjacent stencil cell), and every subsequent adaptive RKF45 substep and stage evaluates the cached polynomials alone — within a step the right-hand side touches neither the grid nor the finite differences. The fit is node-exact on the 27 stencil nodes and, for smooth fields, locally third-order accurate, $\mathcal{O}(\Delta x^3)$, while the finite-difference error of the stencil values is $\mathcal{O}(\Delta x^6)$ and does not limit the convergence. A degenerate or non-finite fit drops the particle via the loss policy (Sec. 4); there is no fallback interpolant in this mode. The sampling points are clipped to the cells interior to the local processor box, so the stencil never reaches the ghost region and the sampler is safe under MPI domain decomposition; in a multi-process run a particle sitting on, or within a few cells of, a rank face has its stencil centre clipped away from it, so the one-sided 27-point fit fails its validity check and the particle is dropped at the first step (the other back-ends are unaffected) — with the default 2×2 (y, z) decomposition of a 4-rank unigrid run every equatorial particle lies on the $z = 0$ rank face, which is why the testsuite runs `l27` single-process.

local27 (default). A self-contained variant of `l27` with a wider stencil and a higher-degree fit: for each of the three stored time levels the thorn fits, for each metric component and each nodal Christoffel symbol, the 125-node $(5 \times 5 \times 5)$ quartic tensor-product polynomial through the stencil surrounding the particle (three separable 5×5 inverse-Vandermonde transforms, the 125×125 analogue of the `l27` matrix), and blends the three level fits with a three-node (quadratic) Lagrange interpolation in the particle's coordinate time. The fit is node-exact on all 125 grid nodes and formally fifth-order accurate in space. It uses no external interpolator thorn. The fits are cached per particle as in `l27`. Two degradation paths keep the sampler well defined on AMR meshes and in strong curvature: when a past time level is not yet filled (the first step, or a freshly grown refinement region) the time blend degrades to a two-level linear blend or to the current level only; and when the nodal Christoffel samples are non-finite (a sixth-order finite-difference window touching the excised or singular region) the Christoffel symbols are instead assembled analytically from the metric fits and their spatial derivatives.

aei (fourth-order pointwise). `metric_interp = aei` evaluates, at the particle position, a fourth-order pointwise Lagrange interpolant of the ten 3+1 fields (lapse, three shifts, six spatial-metric components) and of their first spatial derivatives, using the `AEILocalInterp` thorn [Einstein Toolkit, 2026a]; the metric and its Christoffel symbols are then composed pointwise from these sampled values. The two most recent stored time levels are blended linearly in the particle's coordinate time, and $\partial_t g_{\mu\nu}$ is the first-order difference $(L_0 - L_1)/\Delta t$. Unlike `l27` and `local27`, no per-particle fit is cached: the interpolant is re-evaluated at every RKF45 stage.

carpet. `metric_interp = carpet` is a pointwise sampler with the same fourth-order `AEILocalInterp` spatial molecule as `aei`, evaluated on the current time level; the time derivative is the second-order three-point difference $(3L_0 - 4L_1 + L_2)/(2\,\Delta t)$ of the full 4-metric, the convention of the `CarpetInterp` driver [Einstein Toolkit, 2026b] after which the sampler is named. Like `aei`, it runs unchanged under OpenMP and the owner-computes MPI mode.

4 Implementation

`Geodesic` is written in C as a Cactus/Einstein Toolkit thorn. The particle state $(t, x, y, z, u^t, u^x, u^y, u^z)$ is stored in Cactus interface arrays `particle_tt`, `particle_tx`..., and the integration loop is a plain C loop over the particles, wrapped in an OpenMP `parallel for` with `schedule(dynamic,1)`:

```
#pragma omp parallel for schedule(dynamic,1)
for (mi = 0; mi < particle_n_total; mi++) {
    /* per-particle RKF45 step on the locally evaluated metric */
}
```

There is no shared state inside the loop: each thread evaluates the metric for its own particle (either analytically, Sec. 3.1, or from the grid with one of the local samplers of Sec. 3.2) and advances its own Runge–Kutta state. The only synchronization within one rank is the OpenMP barrier at the end of the parallel region; in multi-process runs it is followed by the MPI state exchanges described below.

MPI and AMR. The particle arrays are declared `DISTRIB=CONSTANT` (replicated on every rank) and are checkpointable, so the particle state survives a restart. In a multi-process grid run the domain is partitioned among the ranks and each rank integrates the particles owned by its local patch (interior-box ownership with a claim margin). A fast particle can cross a rank face within a single Cactus step, so in this mode the step is integrated in coordinate-time chunks: the chunk length is chosen such that every live particle stays strictly inside its owner's interior box, and at each chunk boundary the full state is exchanged and the owner map is recomputed, so that a particle that has crossed into a neighbour's box is picked up by that rank for the next chunk. Every rank derives the identical chunk boundaries from the identical replicated state, so agreeing on the chunking needs no extra communication. After the last chunk and the loss policy, an `MPI_Allgather` exchanges the per-particle state, so the replicated arrays are bit-identical on every rank. This owner-computes scheme reproduces the single-process baselines bit-for-bit in the convergence series (Sec. 5.3); it currently supports up to four ranks (a 2×2 decomposition in y and z, one patch per rank per level). In multi-process `Exact` runs the analytic metric is valid everywhere, so rank 0 integrates all particles and writes the dump. The integration is scheduled at `CCTK_POSTSTEP` in Carpet's level mode: each call acquires the calling level's grid pointers and geometry into a per-level descriptor, and the particles are integrated once per call batch — under Carpet's time subcycling, on the finest-level call, with the step given by the elapsed coordinate time — so the integrator works unchanged in an AMR evolution. Each particle is integrated on the finest refinement level whose safe region contains it (up to 16 levels, `geodesic_amr_levels`, default 3), and MultiPatch (Thornburg [Thornburg, 2004]) block systems are supported: a particle that crosses a block boundary has its position and spatial velocity transformed by the block Jacobians.

Particle lifecycle. What happens to a particle that leaves the computational domain or whose four-velocity fails the normalization check (no future-directed root of Eq. (3)) is selected at run time with `particle_loss_policy`: `drop` (the default) marks the particle state as NaN and stops tracking it, `flag` freezes it at its last valid state, and `respawn` re-places it at a uniformly random position inside the safe box, the grid domain inset by seven cells, subject to the rejections $r <$ excised_radius $+ 1.5$ (inside the exclusion sphere), $r >$ initial_radius (outside the initial particle shell), and $|z| <$ particle_middle_sp (within `particle_middle_sp` of the midplane, default 1.0), with $u^i = 0$ and a u^t placeholder; the per-step re-solve of Eq. (3) then normalizes the four-velocity

on the next step. Every loss is reported to the log with the particle index. The scaling run of Sec. 6 uses `respawn`, which keeps the particle count constant over long runs and prevents a single lost particle from poisoning the statistics.

Diagnostics. With `particle_dump_every = k > 0` the thorn writes the full particle state $(t, x, y, z, u^t, u^x, u^y, u^z, \tau)$ to the file named by the parameter `particle_dump_file` (default `geodesic_particles.tx` the file begins with the initial state ($t = 0$) recorded before the first integration step, followed by one block of rows every k iterations (rank 0 only); this file is the data source for all verification figures below. `rverbose = yes` prints the particle state at every evaluation of the ODE right-hand side to the log for debugging.

5 Verification

5.1 Test orbits

All verification runs use $M = 2$, $a = 0.5$ (the values of the `KerrSchild` initial data [Moreno et al., 2002]) and a 3+1 grid with Cartesian cells of size Δx. Three bound orbits are used (Table 1), each generated and validated independently in Python before the thorn run:

Table 1: Test orbits used for verification ($M = 2$, $a = 0.5$). u^t and u^x, u^y, u^z are the four-velocity components in the par file. $\Delta E/E$ is the value measured by the independent Python reference over the same interval.

orbit	initial (x, y, z)	u^t	u^x, u^y, u^z	radial range	ref. $\Delta E/E$
P0 circular ($r = 10$)	$(10, 0, 0)$	1.53428	(0, 0.67238, 0)	$r = 10$	1.2×10^{-9}
P1 tilted	$(-13.448, 1.180, 0)$	1.32221	$(-0.01827, -0.50381, 0.1)$	[12.86, 16.88]	5.0×10^{-9}
P2 eccentric	$(15, 0, 0)$	1.27607	(0, 0.45176, 0)	[12.03, 15]	5.3×10^{-9}

P0 is a circular orbit of radius $r = 10$ (azimuthal period $T_\phi = 143.4\,M$); P2 is an eccentric orbit with apocenter $r_a = 15$ and pericenter $r_p = 12.03$; P1 is P2's orbit given a small vertical kick at one point, tilting it out of the equatorial plane. Figure 1 shows the three orbits as integrated by the thorn (Exact mode, $T = 1000\,M$): P0 closes on itself in the equatorial plane, while the rosette patterns of P1 and P2 reflect the relativistic periastron advance of the Kerr spacetime.

5.2 Conservation of E, L_z and the norm

We integrate the three orbits simultaneously with `Exact = yes` for $T = 1000\,M$ (10^4 Cactus steps of $\Delta t = 0.1\,M$, i.e. about seven azimuthal periods of P0), on a 100^3 grid. Figure 2 shows the relative drift of E and L_z and the norm residual $|g_{\mu\nu}u^\mu u^\nu + 1|$ for all three orbits. The relative energy drift stays at the level of the independent reference integration ($\leq 2.8 \times 10^{-10}$) and the norm is held at the roundoff level ($\leq 8.9 \times 10^{-16}$) throughout, confirming that the RKF45 integrator with per-step renormalization reproduces the geodesic equation accurately over many orbital periods.

5.3 Grid convergence of the interpolation mode

To quantify the accuracy of the default `local27` sampler, we track the P0 circular orbit ($M = 2$, $a = 0.5$, $r = 10$, azimuthal period $T_\phi \simeq 143.4\,M$) for just over one full azimuthal period ($t = 150$) on grids of 50^3, 100^3 and 200^3 cells ($\Delta x = 1.0, 0.5, 0.25$) with `Exact = no`, and compare the final

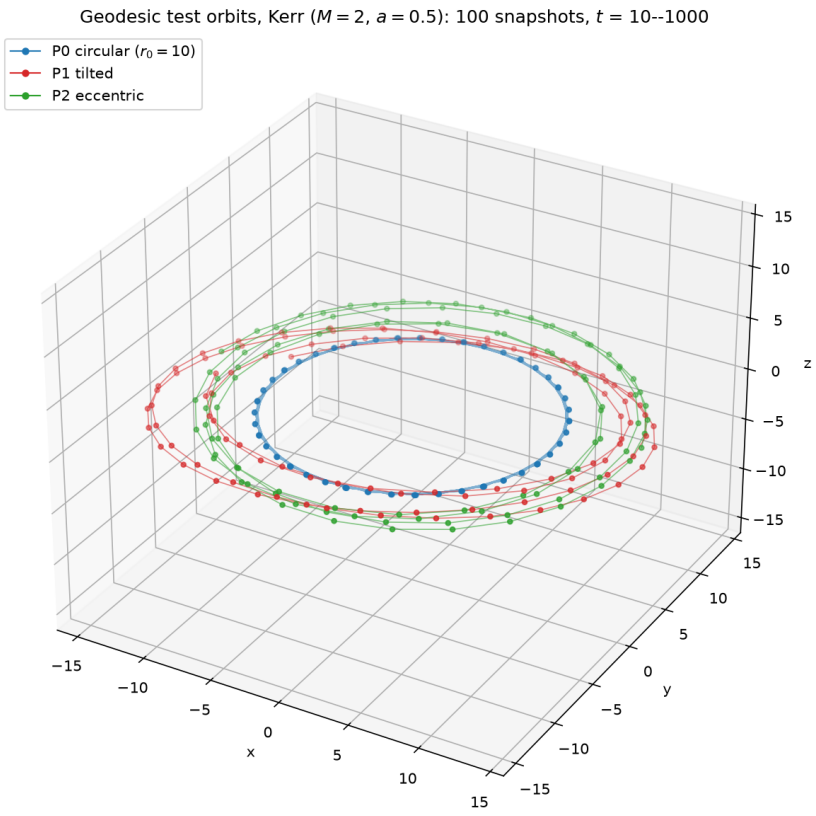


Figure 1: The three test orbits of Table 1 in the Kerr spacetime ($M = 2$, $a = 0.5$), integrated by the thorn in Exact mode over $T = 1000\,M$ (100 snapshots). P0, a circular orbit of radius $r = 10$ in the equatorial plane (blue); P1, P2's orbit given a small vertical kick, precessing out of the plane (red); P2, an eccentric orbit with $r_a = 15$, $r_p = 12.03$ (green).

particle position with an independent Python RK4 integration of the exact metric (step $0.02\,M$). Figure 3 shows the orbit traces and the final-position error versus grid spacing: the error decreases from 2.3×10^{-2} to 4.4×10^{-4} and 9.5×10^{-6} on the two refinements (factors of 52 and 46), a measured local convergence order of 5.6, and the exact-metric run agrees with the reference to 1.7×10^{-6}, which is the level at which two independent integrators differ. The quartic fit used by `local27` is formally fifth-order accurate in space; the measured rate of 5.6 on this orbit is consistent with that formal order, and we report the measured rate as such.

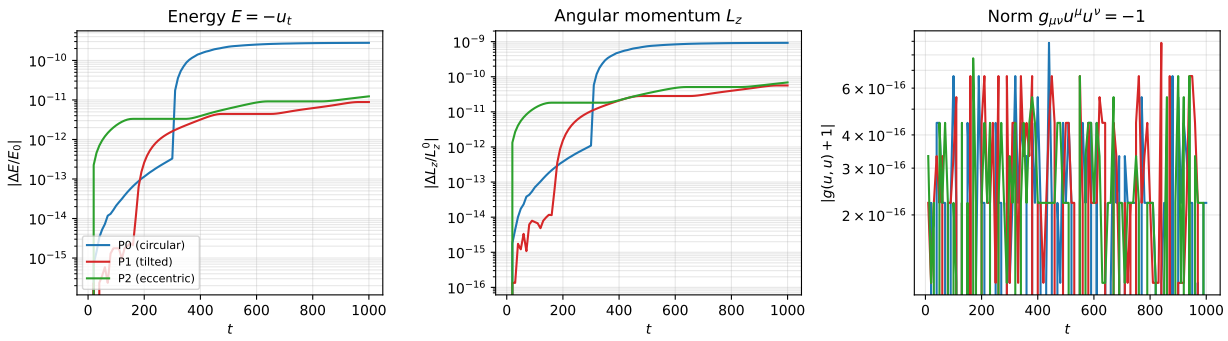


Figure 2: Conservation over $T = 1000\,M$ (Exact mode): relative drift of the energy E (left) and axial angular momentum L_z (middle), and the norm residual $|g_{\mu\nu}u^\mu u^\nu + 1|$ (right), for the three orbits of Table 1. All quantities are conserved at the $\lesssim 10^{-9}$ level.

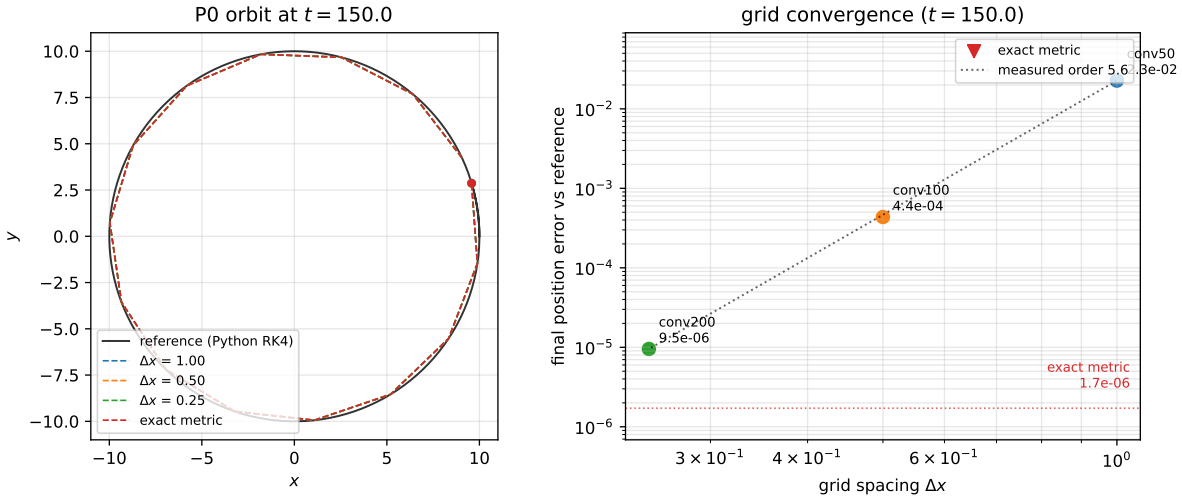


Figure 3: Grid convergence of the P0 circular orbit ($M = 2$, $a = 0.5$, $r = 10$; $t = 0 \ldots 150$, just over one full period). Left: orbit traces in the x–y plane for the interpolation runs ($\Delta x = 1.0, 0.5, 0.25$) and the exact-metric run, compared with the independent Python reference (solid). Right: final-position error versus grid spacing (markers), with a reference line of the measured order 5.6 (dotted) and the exact-metric error level (dotted horizontal line).

5.4 Comparison of the grid samplers on a precessing orbit

To compare the grid samplers on an orbit closer to production use, we track 100 particles that all move on the same bound equatorial Kerr ellipse ($M = 2$, $a = 0.5$, pericentre $r_p = 16$, apocentre $r_a = 64$, $E = 0.9761517$, $L_z = 8.1632196$), spread over the orbit, on the static Kerr–Schild field written onto a 144^3 base grid ($\Delta x = 1$) with three nested refinement boxes ($\Delta x = 0.5$, 0.25, 0.125 out to radii 36, 18, 9). The ellipse precesses by 117.5° (0.6528π) per radial period ($T_r = 1349.7$), so the orbit stays inside the computational domain ($16 \le r \le 64$; domain half-size 72) for the whole run, $t = 0 \ldots 3200\,M$ (about 2.37 radial periods); the particle state is dumped every 10 time units, and a particle is counted as lost when it leaves the safe region of its grid patch or fails an ODE step

(the thorn's drop policy). The six runs are `Exact = yes` (reference, single process, 20 OpenMP threads), `metric_interp = aei` and `metric_interp = carpet` (each single process, 20 threads; both evaluate the same AEI order-4 pointwise Lagrange interpolant and differ only in the data-acquisition path), and `metric_interp = local27` in three configurations: the default four-process mode (four processes, 20 threads each), the explicit `local27` four-process mode (four processes, 20 threads each), and a single-process control (one process, 20 threads).

Table 2: Comparison of the grid samplers on the precessing ellipse of Sec. 5.4 (100 particles, $t = 0 \ldots 3200\,M$; no particle lost in any run). ΔE, ΔL_z: maximum and mean absolute deviation of the conserved quantities from their initial values, over the run. "p0": radial position of particle 0 at the end of the run.

metric evaluation	run	lost	survivors	$\max\lvert\Delta E\rvert$ / mean	$\max\lvert\Delta L_z\rvert$ / mean	p0
exact (reference)	1 process, 20 threads	0	100	—	—	$r = 32.47$ (alive)
aei	1 process, 20 threads	0	100	1.14×10^{-6} / 1.38×10^{-7}	6.53×10^{-5} / 1.97×10^{-5}	$r = 32.47$ (alive)
carpet	1 process	0	100	(identical to `aei`)	(identical to `aei`)	$r = 32.47$ (alive)
local27 (explicit)	4 processes, 20 threads ea.	0	100	6.41×10^{-5} / 1.76×10^{-5}	1.15×10^{-3} / 5.56×10^{-4}	$r = 32.45$ (alive)
local27 (default)	4 processes, 20 threads ea.	0	100	(identical to explicit)	(identical to explicit)	$r = 32.45$ (alive)
local27 (control)	1 process, 20 threads	0	100	1.07×10^{-5} / 2.01×10^{-6}	6.92×10^{-4} / 3.44×10^{-4}	$r = 32.44$ (alive)

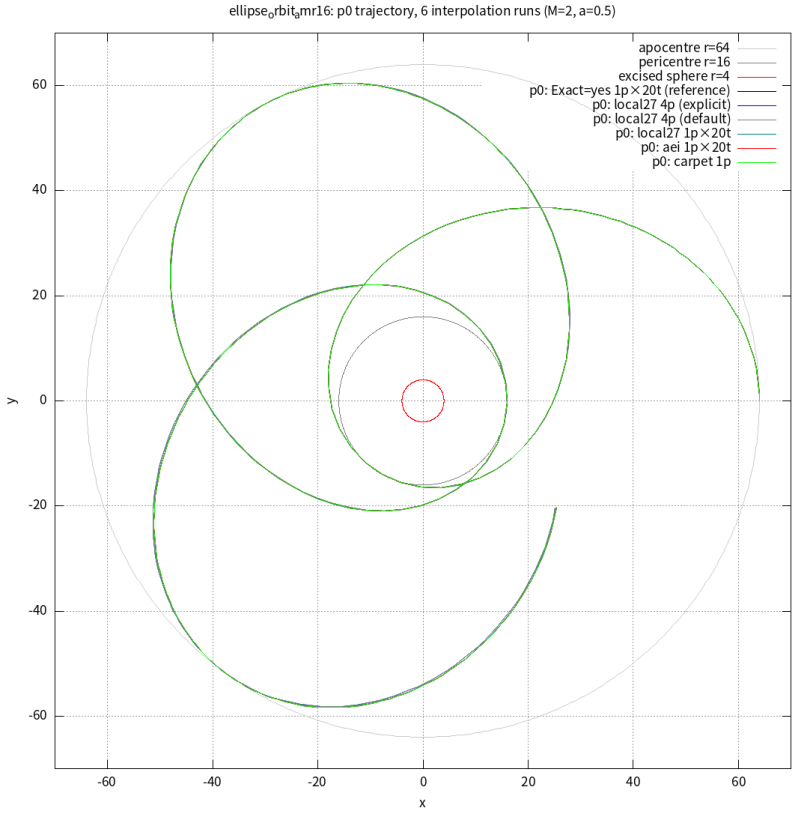


Figure 4: Particle 0 of the precessing-ellipse comparison (Sec. 5.4) in the x–y plane: exact reference (black), the two `local27` four-process runs (blue solid and gray dashed, bit-identical to each other), the `local27` single-process control (teal dashed), `aei` (red) and `carpet` (green; bit-identical to `aei` in this test). Circles: apocentre ($r = 64$), pericentre ($r = 16$) and the excision sphere ($r = 4$, dashed).

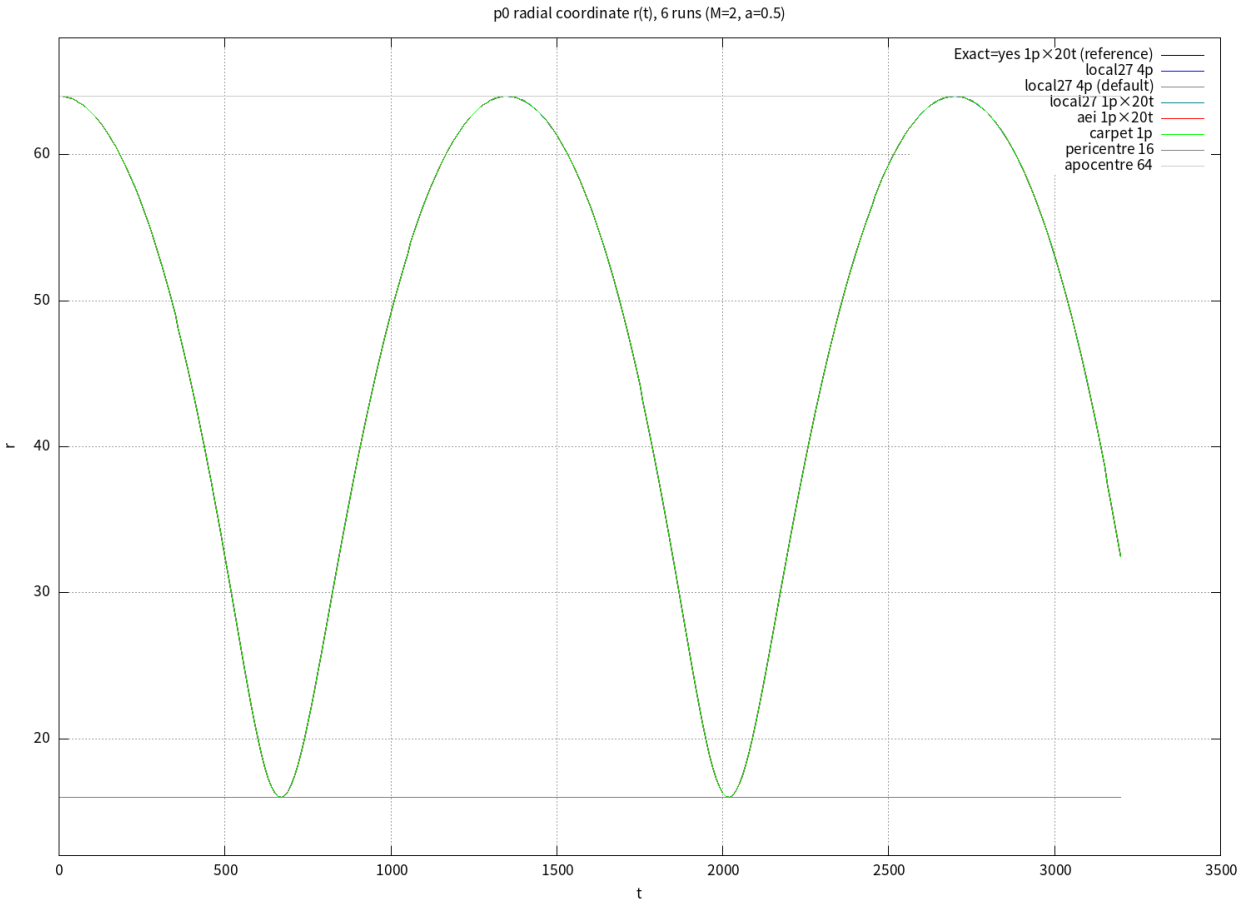


Figure 5: Radial coordinate $r(t)$ of particle 0 for the runs of Table 2. Dashed: pericentre ($r = 16$) and apocentre ($r = 64$) of the reference ellipse.

No particle is lost in any of the six runs: all 100 particles complete the full $t = 0 \ldots 3200$ run without leaving the safe region of their grid patch or failing an ODE step. The deviation from the exact-metric reference separates the runs cleanly (Table 2, Figure 5): the two single-process AEI order-4 runs (`aei`, `carpet`; bit-identical to each other) track the reference to 2.1×10^{-2}, the single-process `local27` control to 2.3×10^{-1}, and the two four-process `local27` runs (bit-identical to each other) to $O(1)$, peaking at 2.54 near $t \simeq 2680$ and decaying to 1.41 by $t = 3200$. In the single-process runs the dominant error is the grid's own coherent discretization of the exact Kerr–Schild background, which all three samplers reproduce at the $10^{-2} \ldots 10^{-1}$ level; the node-exact `local27` fits carry it further than the AEI order-4 interpolant (2.3×10^{-1} versus 2.1×10^{-2}). The $O(1)$ deviation of the four-process runs — absent in the single-process control of the same fit — is the owner-computes multi-process limitation documented in Sec. 4 for particles crossing rank and patch boundaries; a particle that nevertheless drifts out of its owner's box within a chunk is flagged and handed to the loss policy instead of letting the run corrupt. This has no adverse consequence in production use: in a full `GRHydro` evolution or any grid-carried spacetime the grid field *is* the true metric, and the single-process runs show that the trackers follow it faithfully.

Run notes. The default-mode four-process run and the explicit `local27` four-process run are bit-identical to each other (identical mode, different par spelling), confirming the default path of

Sec. 3.2. `aei` and `carpet` were run with a single process in this comparison; that is a choice of convenience, not a limitation — both back-ends pass their four-rank testsuite runs (`conv100_aei`, `conv100_carpet`), and it is the `l27` path that must be run single-process (Sec. 3.2).

5.5 Geodesic orbits on an AMR mesh

The grid-mode samplers read the metric from the refinement level that contains the particle, which makes them usable on Carpet's AMR meshes without modification. We verify this on the same circular orbit as the unigrid reference `conv100.par` (Kerr–Schild, $M = 2$, $a = 0.5$, particle on the $r = 10$ equatorial circle, integrated to $t = 150.1$, ~ 1.05 orbital periods, $T \simeq 143$), now evolved on an AMR mesh with three levels of resolution: a 50^3 base grid ($\Delta x = 1$) refined twice (factor 2) by a box centred on the origin (half-width 16 cells on the first refinement level, then 8), so that the finest level has $\Delta x = 1/4$. The orbiting particle lies on the first refinement level ($\Delta x = 1/2$) for its whole path — the same local resolution as the 100^3 unigrid reference — and crosses onto the second level ($\Delta x = 1/4$) near the box diagonals. The particle integrator is otherwise unchanged (RKF45, same seed and initial data as the reference run), so the only differences between the two set-ups are the grid structure and the driver's 4:1 subcycling.

Table 3: AMR (3 levels) vs. unigrid reference (conv100): maxima over the 16 dump times $t = 0.1, 10.1, \ldots, 150.1$ of the Cartesian position difference $\max_i |\Delta x^i|$, the radial difference $|\Delta r|$, and the orbital phase difference $|\Delta\phi|$. The reference run leaves `metric_interp` unset and so uses the default `local27`; each row is that sampler's AMR-to-unigrid difference, not a ranking of sampler accuracy.

sampler	$\max_i \lvert\Delta x^i\rvert$	$\max \lvert\Delta r\rvert$	$\max \lvert\Delta\phi\rvert$ (rad)
local27	1.2×10^{-1}	4.3×10^{-2}	1.3×10^{-2}
aei	1.4×10^{-1}	5.0×10^{-2}	1.6×10^{-2}
carpet	3.4×10^{-1}	2.3×10^{-1}	4.2×10^{-2}

All three samplers keep the particle on a stable orbit throughout the integration (no blow-up, no lost particle). `local27` and `aei` stay within 5×10^{-2} (0.5%) of the reference in radius and 1.7×10^{-2} rad ($\sim 1°$) in phase; the `carpet` sampler drifts to 2.3×10^{-1} (2.3%) in radius, the largest of the three on this mesh. The reference run (`conv100.par`) leaves `metric_interp` unset and so uses the default `local27`: the `local27` row measures the effect of the mesh alone, whereas the `aei` and `carpet` rows include a sampler change as well, so this table does not rank the samplers by accuracy (the comparison against the exact metric is Sec. 5.4). At the first dump after the mesh settles ($t = 10.1$) all three deviations are already at their small-t level ($\lesssim 5 \times 10^{-3}$ in position), so no transient is associated with the refinement. The pointwise samplers therefore resolve the metric correctly on an AMR mesh, including across the level boundaries, and the geodesic equation can be integrated in a Carpet AMR evolution (with its 4:1 subcycling) without any modification of the integrator.

5.6 Robustness on a two-puncture merger background

The checks above all use a single-black-hole Kerr–Schild field, where the dominant error source is the grid's own discretization. A closer production use case is the field of a binary black hole, whose geodesic structure is itself of observational interest [Chernov, 2024]. We therefore place the tracker

on the two-puncture metric of the `GW150914` binary (the first directly detected gravitational-wave signal [LIGO Scientific Collaboration and Virgo Collaboration, 2016]; component masses 36 and $29\,M_\odot$, stored in the par files as $m_+ = 36/65$ and $m_- = 29/65$, i.e. normalized to total mass $M = 1$), built by `TwoPunctures` with the GW150914 spin and kick parameters on a 144^3 Cartesian grid ($\Delta x = 0.25$, domain ± 18) with the puncture excision radius set to 2. There is no analytic reference orbit on this background, so the relevant checks are well-posedness and stability rather than absolute accuracy.

An important caveat: the raw `TwoPunctures` initial data is not the merger metric itself. Once the field is evolved with `ML_BSSN`, it relaxes and differs significantly from the frozen initial geometry. We therefore make the time-dependent, `ML_BSSN`-evolved background the main robustness test, and treat the frozen initial data only as a secondary stability check.

In all runs we place 100 test particles on two initial rings, each run with the three grid-sampling back-ends (`local27`, `aei`, `carpet`; the latter two sample through the `AEILocalInterp`/`LocalInterp` interface and are bit-identical to each other): a *far* ring at $r = 10$ (prograde tangential velocity 0.32, inward radial 0.10) and a *near* ring at $r = 8$ with a strong inward radial velocity (0.960), so that the particles spiral into the two punctures.

Time-dependent background. With `ML_BSSN` evolution active on the unigrid two-puncture field (timestep 0.05, at the stability limit of that scheme at $\Delta x = 0.25$), the near-ring trajectories visibly depart from the frozen-geometry ones: the two runs agree to $\lesssim 0.13$ in position through $t \simeq 4$ and diverge to ~ 0.8 by $t \simeq 7 \ldots 8$, with 73 of the 100 particles surviving to $t = 10$ (against 78 in the frozen run). In the 400-step version of this run the coarse unigrid background itself develops `NaN` at a puncture just after the $t = 10$ snapshot, near particle time 10.55 — the resolution limit of that grid — and the `NaN` checker terminates the run gracefully, an outcome impossible on a frozen geometry and direct evidence that the metric was evolving. Figure 6 shows the `local27` trajectories of that run. The far ring is integrated on the same evolved background over 100 steps ($t = 1 \ldots 5.95$): all 100 particles remain finite with zero drops, the ring spiraling gently inward from $r \simeq 9.9$ to $r \simeq 9.6$. Finally, the strong-inward near ring is run on the two-puncture `GW150914` background evolved with `ML_BSSN` (8th-order finite differences, 1+log slicing, Γ-driver shift) on a multi-patch AMR mesh with two static-box refinement structures (seven levels each, regridded every 128 steps), over 2048 finest-level steps ($t \simeq 0 \ldots 4.4$): all 100 particles remain finite throughout, with zero drops, on both the `local27` and `aei` back-ends, which agree to 1.0×10^{-1} in position (the grid-discretization level). Figure 7 shows the `local27` trajectories of that run. The tracker is therefore stable in a genuinely time-dependent merger spacetime.

Frozen initial-data check. On the frozen raw initial data (a pure geometry-routing/stability check, not representative of the merger), the far ring completes the full 80-unit run on all three back-ends with zero `NaN` states, and the `local27` trajectory agrees with the `aei`/`carpet` ones to within $\sim 6 \times 10^{-4}$ in position; on the near ring, deterministic `NaN` drops (`loss_policy = drop`) set in from $t = 6$ and reach exactly 50 of the 100 particles by $t = 20$, with an almost identical drop pattern on all three back-ends: `aei` and `carpet` agree bit for bit, and `local27` differs in only two of the fifty dropped particles. Every drop is a physical excursion into the strong-field/excised region rather than a numerical failure of the integrator: on a Kerr background these orbits are the plunging geodesics whose end states are known analytically [Dyson and van de Meent, 2023]. All six frozen runs (near and far ring, three back-ends each) are part of the thorn's automatic validation suite (`test/test.ccl`); the NaN-aware test comparator turns the deterministic drop pattern of the near runs into a regression test.

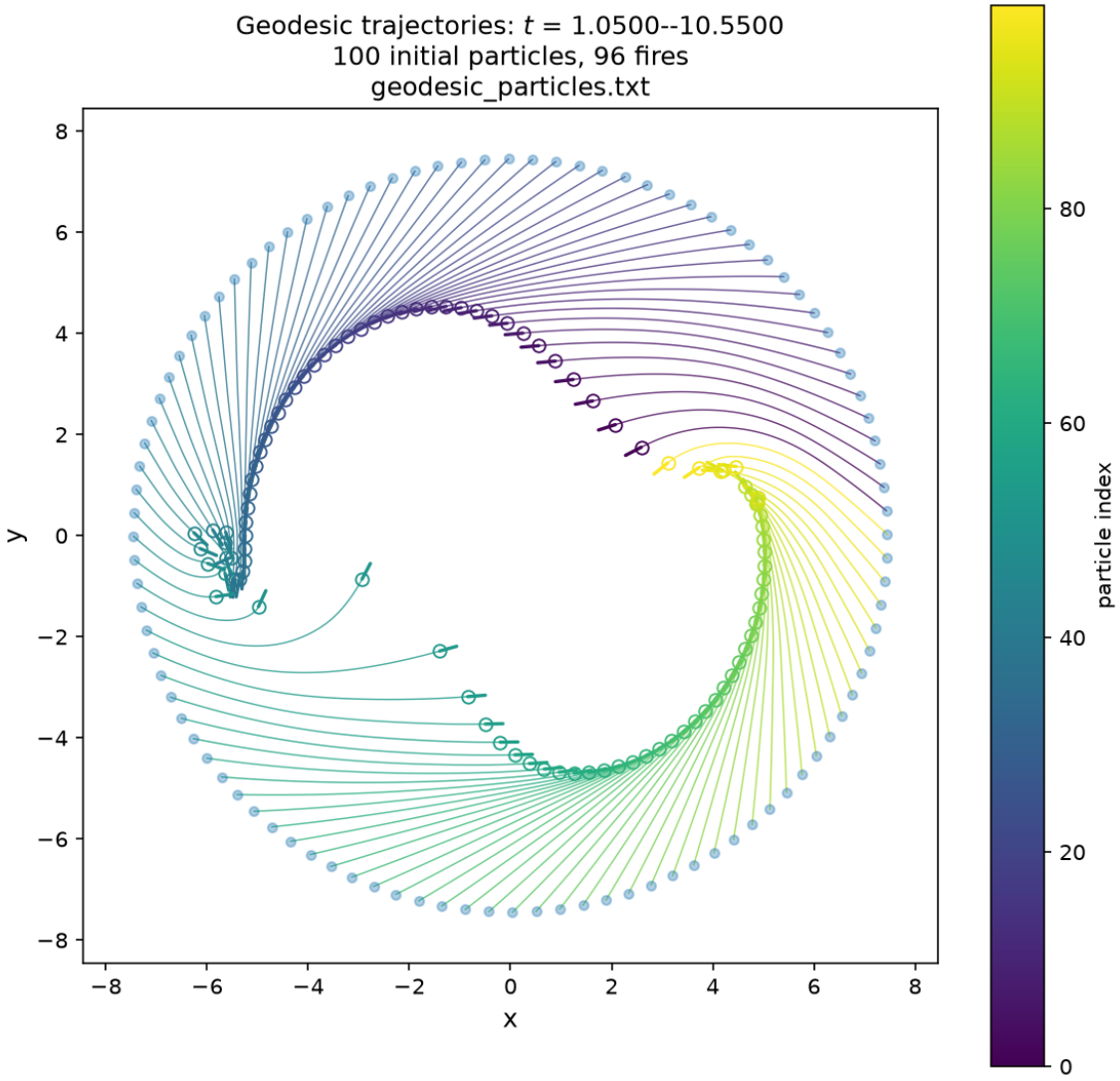


Figure 6: Strong-inward near ring (100 particles, `local27`) on the `ML_BSSN`-evolved two-puncture `GW150914` background ($\Delta x = 0.25$ unigrid, $\Delta t = 0.05$), $t = 1.05 \ldots 10.55$. The trajectories clearly depart from the frozen-initial-data geometry. The run ends when the coarse background itself develops `NaN` at a puncture just after $t = 10$ (the resolution limit of that grid); 73 of the 100 particles survive to $t = 10$. Markers are individual snapshots (colour = particle index).

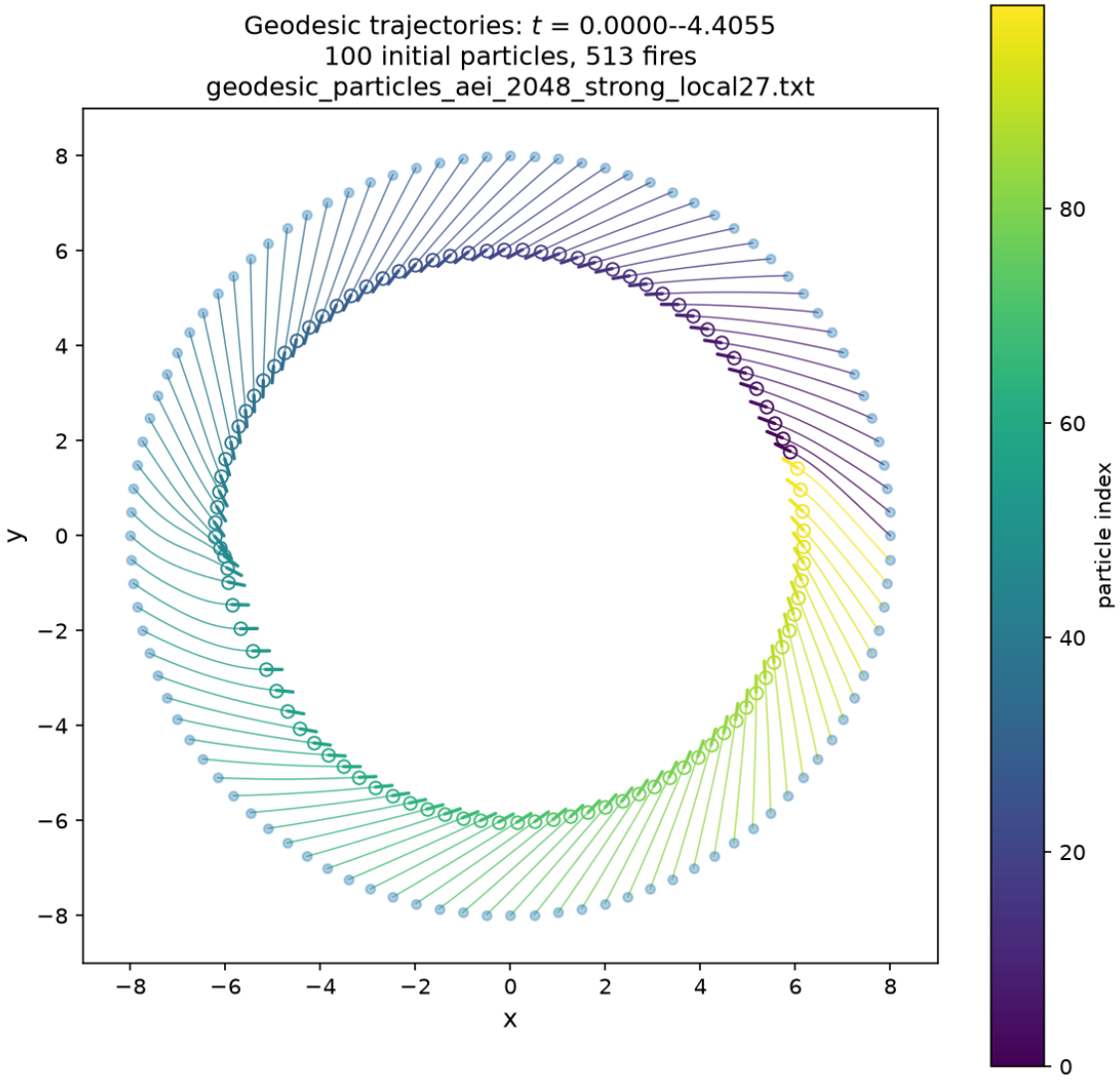


Figure 7: Strong-inward near ring (100 particles, `local27`) on the two-puncture `GW150914` background evolved with `ML_BSSN` on a multi-patch AMR mesh with two static-box refinement structures, over $t \simeq 0 \ldots 4.4$ (2048 finest-level steps). All 100 particles remain finite throughout (zero drops) and wrap around the two punctures as the background evolves; the `aei` back-end agrees with `local27` to 1.0×10^{-1} in position.

6 Strong scaling

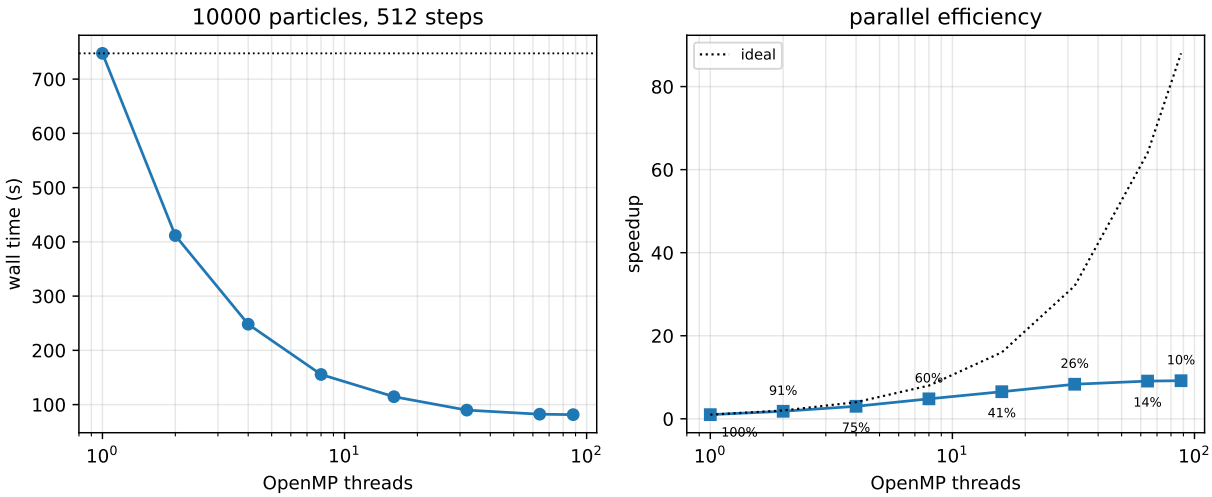


Figure 8: Strong scaling with 10^4 particles, 50^3 grid, 512 steps, `l27` sampler. Left: wall time versus number of OpenMP threads. Right: speedup versus thread count (solid line, ideal) with the parallel efficiency annotated at each point.

The per-particle independence of the integration loop (Sec. 4) should translate directly into parallel speedup. We run a configuration with 10^4 particles (the three orbits of Table 1 plus 9997 particles with random positions inside the safe box and zero velocity, fixed random seed), 50^3 grid, `Exact = no`, 512 steps, on a two-socket node with 44 physical cores (88 logical threads via hyperthreading), varying only `OMP_NUM_THREADS` from 1 to 88. Figure 8 shows the wall time and the parallel efficiency. The speedup grows with the thread count up to 32 threads (8.3×) and then plateaus at 9.2× (efficiency 10.4%) at 88 threads. A per-phase timing breakdown (Cactus `TimerReport`) attributes the plateau to two effects. First, the surrounding Cactus framework (metric bookkeeping, boundary conditions, NaN checks) consumes ∼40 ms per step, nearly independent of the thread count (19.0 s total at one thread versus 22.9 s at 88): about 3% of the ∼1.45 s single-thread step but about 29% of the ∼0.15 s 88-thread step. Second, the parallel particle region itself scales sublinearly: it shrinks from ∼1.41 s to ∼0.104 s per step, a speedup of only 13.7× on 88 threads, consistent with the 44 physical (hyperthreaded) cores, cross-NUMA stencil reads, and OpenMP scheduling overhead.

As a complementary measurement in a production setting, we ran the tracker inside full `GRHydro` evolutions [Font, 2008] with 5000 and with 10^4 particles; each run shared the 88-thread node with three other concurrent instances, i.e. only ∼22 threads were available per instance. The particle integration (`geodesics_integrate` in the `TimerReport`) consumed 49.7 s of 2860.5 s total wall time (1.7%) in the 5000-particle run and 84.1 s of 2749.1 s (3.1%) in the 10^4-particle run: the dominant costs remain the hydrodynamic reconstruction (∼ 27%) and the Riemann solves (∼ 10%), and doubling the particle count grows the integration cost by a factor of only 1.7. Because these fractions were measured under contention from the co-running instances, the overhead in a dedicated run is smaller still: tracking test particles alongside an evolution is essentially free compared to the evolution itself.

7 Conclusion and availability

We have presented `Geodesic`, an Einstein Toolkit thorn that integrates test-particle geodesics in Kerr–Schild Cartesian coordinates. The thorn supports an exact analytic Kerr–Schild metric (grid-independent, verified to be a vacuum solution to beyond machine precision) and a grid mode with four local samplers selected by one parameter (`l27`, `local27`, `aei`, `carpet`), each with its time handling built in. The grid mode assumes no particular analytic form of the metric; for stationary grids $\partial_t g_{\mu\nu} = 0$ is detected automatically, and for time-dependent grids the time derivative is recovered from the stored `ADMBase` time levels (Sec. 3): with `Exact = no`, the thorn tracks geodesics in whatever spacetime the grid carries, stationary or evolving (Sec. 3) — for example the two-puncture `GW150914` binary background of Sec. 5.6. Multi-process runs distribute the particles over MPI ranks with an owner-computes scheme that keeps the replicated state bit-identical on every rank; AMR meshes with up to 16 refinement levels and MultiPatch block systems are supported (Sec. 4). In that mode each particle is an independent unit of work, and an OpenMP parallelization of the particle loop scales to 10^4 particles, reaching a speedup of 9.2× (10.4% efficiency) on the 88 logical threads of a two-socket, 44-core hyperthreaded node. Verification against an independent reference implementation shows conservation of E, L_z and the norm to $\sim 10^{-9}$ over $10^3\,M$, a measured local convergence order of 5.6 of the default sampler, and, on a precessing orbit of 100 particles, agreement with the exact-metric reference at the 2.1×10^{-2} (AEI order-4 path), 2.3×10^{-1} (single-process `local27`) and $O(1)$ (four-process `local27`) levels (Sec. 5.4); and, on a 3-level AMR mesh, orbits of the grid samplers that reproduce the unigrid reference (run with the default sampler) to within 4.3×10^{-2} (`local27`), 5.0×10^{-2} (`aei`) and 2.3×10^{-1} (`carpet`) in radius (Sec. 5.5).

The thorn is released under the GPLv2+ license together with the configuration files used in this work; the code, the configuration files and this paper are publicly available at `https://github.com/wylyhzh/Geodesic`.

Statement on AI assistance

During the preparation of this paper the author used AI assistants (large language models) for debugging the code, drafting and revising the text, and producing the figures. All simulations, data and results are the author's own; every number in the paper was checked by the author, who takes full responsibility for its content.

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